

Style-Aware Retrieval Evaluation Results

Project Goal

The goal of this project is to build a photographer-oriented image retrieval system that combines semantic search with photographic style similarity.

The retrieval pipeline consists of:

1. CLIP semantic retrieval
 2. FAISS nearest-neighbor search
 3. Photographic style-aware reranking
-

Dataset

Dataset: Unsplash Lite

Current subset:

- 1000 downloaded images
- SQLite metadata database
- CLIP embeddings
- Visual descriptors

Extracted style descriptors:

- Brightness
 - Contrast
 - Saturation
 - Warmth
 - Color Histogram
-

Retrieval Pipeline

Stage 1 — Semantic Retrieval

```
Query
↓
CLIP Encoder
↓
512-D Embedding
↓
FAISS Flat Index
```

↓
Top-K Candidates

Stage 2 — Style Reranking

Top-K Candidates
↓
Style Similarity
↓
Combined Score
↓
Final Ranking

Final score:

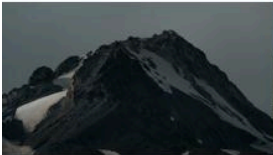
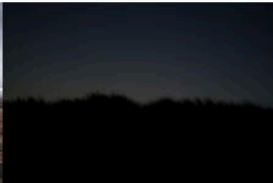
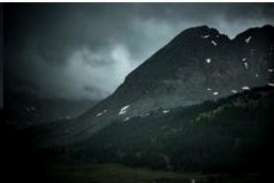
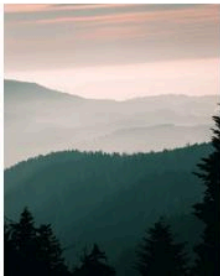
Final Score =
0.75 × Semantic Score +
0.25 × Style Score

Text-to-Image Retrieval Examples

Query: "warm cinematic landscape"

TEXT QUERY

warm cinematic landscape

#1 score=0.2674 YA2KX0JVdE	#2 score=0.2661 dxZfq6Xuz5c	#3 score=0.2654 R2cWLI7yj1o	#4 score=0.2629 cZzC8D5fqIE	#5 score=0.2611 OzD_vkpMM5Y
				
#6 score=0.2607 vDZCV1akWss	#7 score=0.2604 iA0a8ilUFIM	#8 score=0.2594 Xv8nEggOsxw	#9 score=0.2590 xwTgctQbCtA	#10 score=0.2566 utBjiT8TOhA
				

Observations

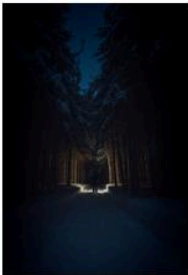
- Landscape images dominate the results.
- Warm orange and yellow tones are common.
- Retrieved images match both semantic and color expectations.

Query: "dark moody forest"

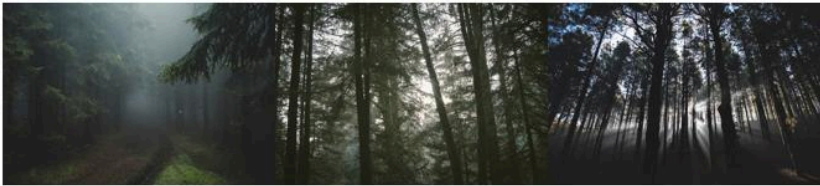
TEXT QUERY

dark moody forest

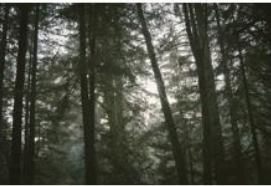
#1 score=0.3052
odfhHtD3igY



#2 score=0.3042
4CDdd1RCt6w



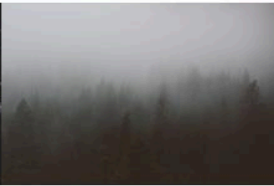
#3 score=0.2979
4qZq7xlqpvM



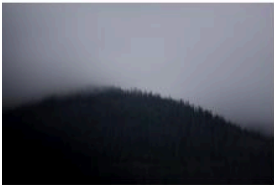
#4 score=0.2967
3DyaiH8xnmY



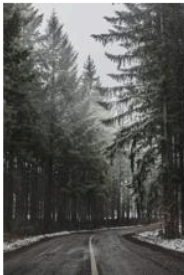
#5 score=0.2949
cv_fqvrkKWs



#6 score=0.2925
_zHYUQmWrzk



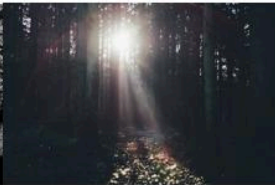
#7 score=0.2922
i3nTy4JcLpY



#8 score=0.2906
qsEJGX4VqYU



#9 score=0.2883
QPg8RkHRIOo



#10 score=0.2842
cM-ZZikRQQs



Observations

- Forest scenes are correctly retrieved.
- Most images have low brightness.
- Dark green and brown tones dominate.

Query: "cold snowy mountain"

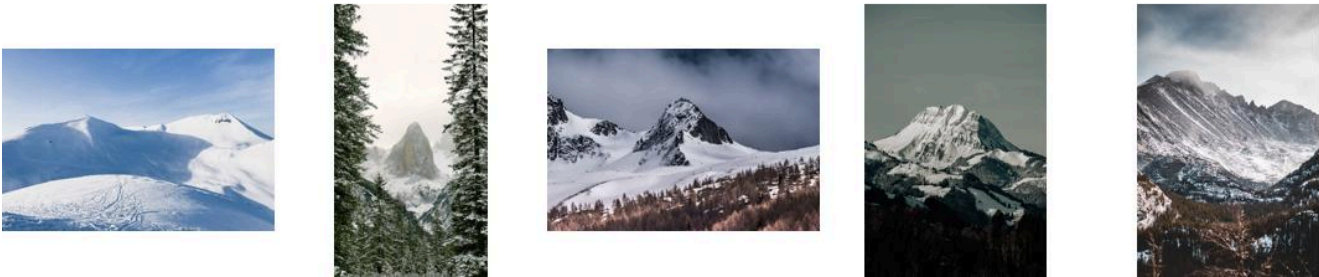
TEXT QUERY

cold snowy mountain

- #1 score=0.2990
rmAK5hu7xsA
- #2 score=0.2988
R2cWLI7yj1o
- #3 score=0.2946
6xTKw5ufGTy
- #4 score=0.2923
fwxVU3WnsFQ
- #5 score=0.2913
_AJwryNo7kE



- #6 score=0.2902
utl-VZuiA4o
- #7 score=0.2886
HL2kVQN85sA
- #8 score=0.2879
hB16dqA6uv4
- #9 score=0.2876
4mhn1oniZql
- #10 score=0.2868
eXUR-Z4MF_A



Observations

- Snow-covered mountain scenes are retrieved.
- Cold blue tones are present.
- High semantic consistency.

Query: "bright tropical beach"

TEXT QUERY

bright tropical beach

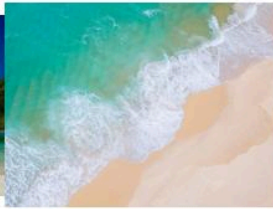
#1 score=0.2850

gweOVL1QjPo



#2 score=0.2835

lrPBS8b4G14



#3 score=0.2820

2opFBNXAJzw



#4 score=0.2817

Qz3el1B2QPs



#5 score=0.2814

7FU42T_XgH8



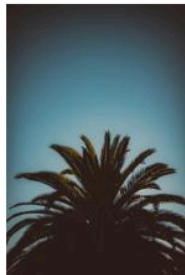
#6 score=0.2792

V63oM8OPJSo



#7 score=0.2782

9_uMcDhylKw



#8 score=0.2777

67sVPJk6Q7l



#9 score=0.2770

Ud9lKknu9c0



#10 score=0.2746

VEco5HQZv00



Observations

- Beaches and oceans dominate the results.
- Bright colors and sunlight are visible.
- Strong match with the text query.

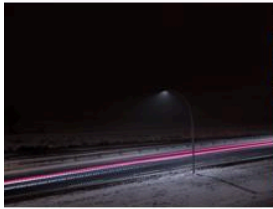
Query: "minimal street photography"

TEXT QUERY

minimal street photography

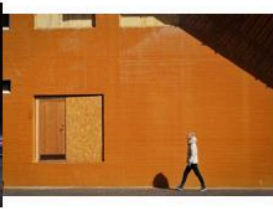
#1 score=0.2794

AvyqZ8XwQ6c



#2 score=0.2764

k_NkqlaRhro



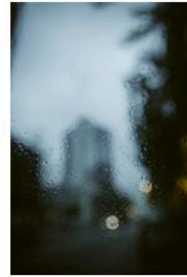
#3 score=0.2685

T5Pm7Zld3sc



#4 score=0.2662

8C1_maRH3nY



#5 score=0.2624

9r-WFdAQQAc



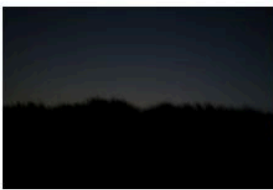
#6 score=0.2624

9_uMcDhylKw



#7 score=0.2622

iA0a8iIUfIM



#8 score=0.2584

i-vgADdh6E4



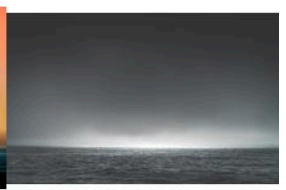
#9 score=0.2579

PzytfAmbOQM



#10 score=0.2559

tLatn4r0nYo



Observations

- Urban scenes are correctly retrieved.
- Composition tends to be simpler.
- Several images resemble minimal photography.

Query: "black and white architecture"

TEXT QUERY

black and white architecture

#1 score=0.2733

qJmfb_wWXhw



#6 score=0.2509

s0xEPfyYjfU



#2 score=0.2616

iRNGsOeKUU0



#7 score=0.2508

EyBKbpPCW7A



#3 score=0.2584

ACt8ycSzpdE



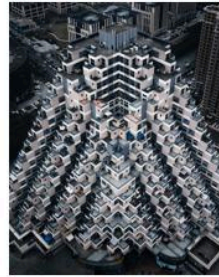
#8 score=0.2494

qsEJGX4VqYU



#4 score=0.2567

Q8x0hSBMMGk



#9 score=0.2491

smjgqsH07JQ



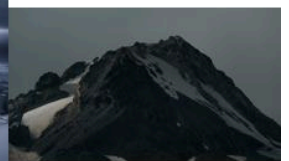
#5 score=0.2537

9wTWfyInJ4Y



#10 score=0.2399

YA12KX0JVdE



Observations

- Architecture-related images are retrieved.
- Several results contain monochrome elements.
- Demonstrates CLIP semantic understanding.

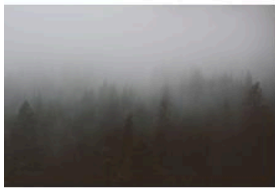
Query: "red cinematic fog"

TEXT QUERY

red cinematic fog

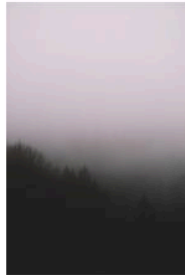
#1 score=0.2749

cv_fqvrkKWs



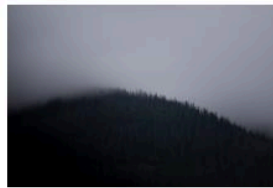
#2 score=0.2724

sfsvTLQh0o4



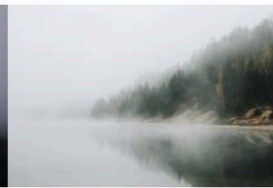
#3 score=0.2683

_zHYUQmWrzk



#4 score=0.2677

S5RBgaziupg



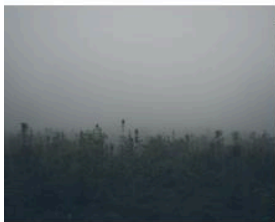
#5 score=0.2675

qsEJGX4VqYU



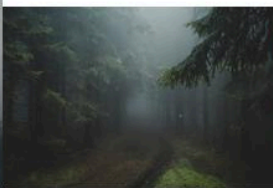
#6 score=0.2659

x9RN-Zlzkf0



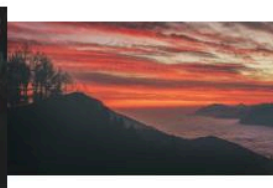
#7 score=0.2652

4CDdd1RCt6w



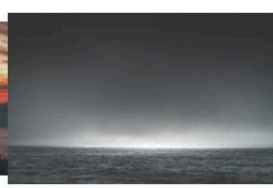
#8 score=0.2646

ilnYwn194Tk



#9 score=0.2606

tLatn4r0nYo



#10 score=0.2582

_AJwryNo7kE



Observations

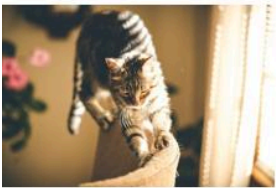
- Red color tones are present.
- Atmospheric scenes dominate.
- Shows CLIP's ability to combine color and mood concepts.

Query: "cat riding a spaceship"

TEXT QUERY

cat riding a spaceship

#1 score=0.2660
o9Z91mK4Q5l



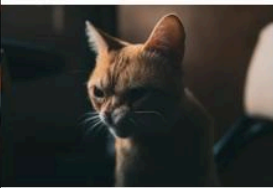
#2 score=0.2546
dt0A75WG_aE



#3 score=0.2455
YUwcEo-zrUA



#4 score=0.2453
5zBI2v4aYXM



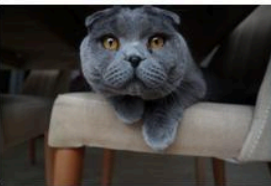
#5 score=0.2452
7ZiG0-YeFJU



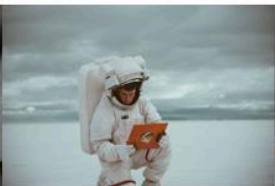
#6 score=0.2409
EVXB_Is-Uql



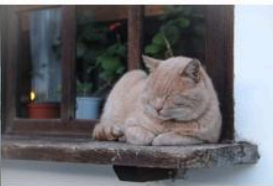
#7 score=0.2408
0lTYt-tkq4l



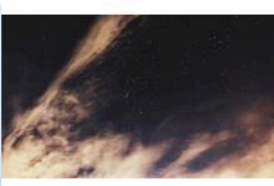
#8 score=0.2356
eqbLoTqQKZs



#9 score=0.2348
B1KFwtFFZi8



#10 score=0.2332
Bwkl85sszNM



Observations

- No exact matches exist in the dataset.
- Results demonstrate CLIP behavior on out-of-distribution queries.
- Retrieved images share partial semantic concepts.

CLIP vs Style-Reranking Comparison

Query Image: 9U_uCvfpptk

QUERY

9U_uCvfpptk.jpg

CLIP ONLY

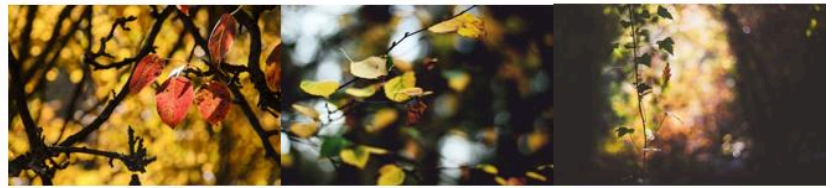
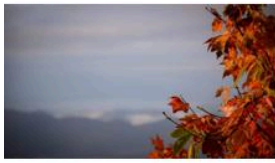
#1 score=0.9062

21r3djZK3Lc
sem=0.9062

#2 score=0.8930

vLv5PSHy4m4
sem=0.8930

#3 score=0.8810

IaUQaRORK6k
sem=0.8810

#4 score=0.8773

pl6XZycqe-k
sem=0.8773

#5 score=0.8681

OIBkP6gOZ7A
sem=0.8681

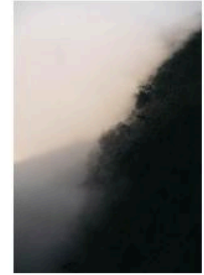
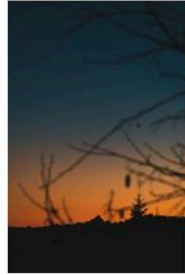
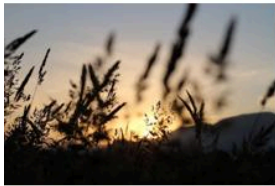
#6 score=0.8671

LUQehBE2FOA
sem=0.8671

#7 score=0.8601

2HLQBF2hZY0
sem=0.8601

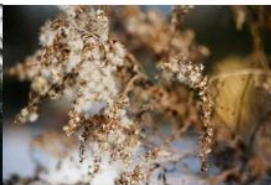
#8 score=0.8562

2EGf-Ag2tyA
sem=0.8562

#9 score=0.8561

Fnb73SxGQcK
sem=0.8561

#10 score=0.8548

YcCTVdhPiw4
sem=0.8548

QUERY

9U_uCvfpptk.jpg

CLIP + STYLE RERANK

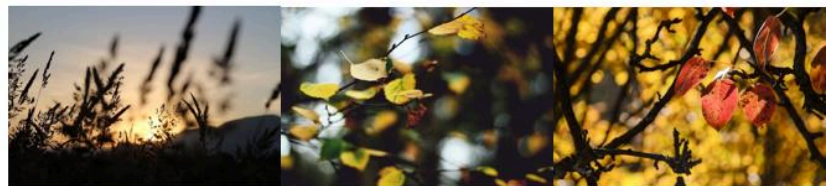
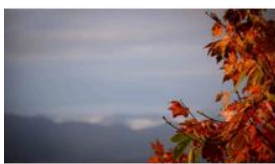
#1 score=0.8830

pl6XZycqe-k
sem=0.8773
style=0.9001
final=0.8830

#2 score=0.8765

vLv5PSHy4m4
sem=0.8930
style=0.8271
final=0.8765

#3 score=0.8723

21r3djZK3Lc
sem=0.9062
style=0.7705
final=0.8723

#4 score=0.8706

YcCTVdhPiw4
sem=0.8548
style=0.9179
final=0.8706

#5 score=0.8689

oenAIDZSfE
sem=0.8481
style=0.9310
final=0.8689

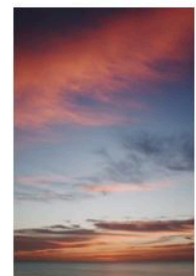
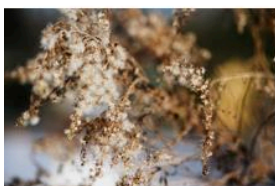
#6 score=0.8624

OIBkP6gOZ7A
sem=0.8681
style=0.8456
final=0.8624

#7 score=0.8613

IaUQaRORK6k
sem=0.8810
style=0.8021
final=0.8613

#8 score=0.8565

O_wOFhmlXtY
sem=0.8473
style=0.8842
final=0.8565

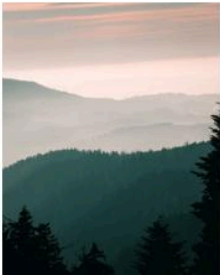
#9 score=0.8520

utBjtT8TohA
sem=0.8510

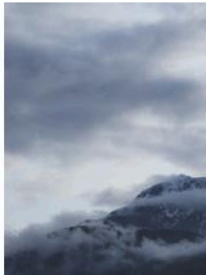
#10 score=0.8516

9hOKBPVEc10
sem=0.8437

style=0.8548
final=0.8520



style=0.8752
final=0.8516



Observations

- CLIP retrieves semantically similar scenes.
- Reranking improves visual consistency.
- Lighting and color temperature become more aligned.

Query Image: 9wTWfYlnJ4Y

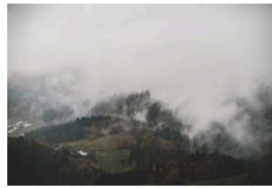
QUERY

9wTWfYlnJ4Y.jpg

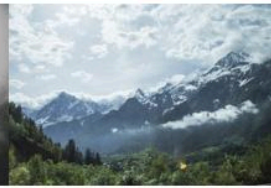


CLIP ONLY

#1 score=0.8292

795KLb6DI7E
sem=0.8292

#2 score=0.8116

rZ2LHT8eU9w
sem=0.8116

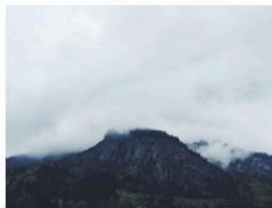
#3 score=0.8062

hB16dqA6uv4
sem=0.8062

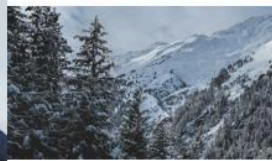
#4 score=0.7884

4mhn1oniZqI
sem=0.7884

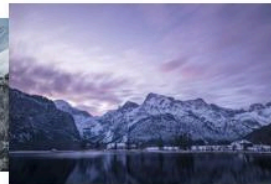
#5 score=0.7873

cDD83wV627U
sem=0.7873

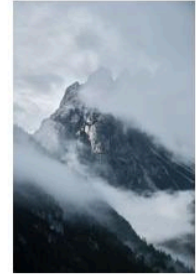
#6 score=0.7803

Ey1CHKvnShc
sem=0.7803

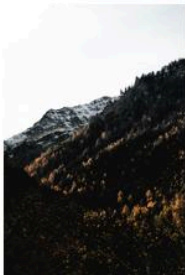
#7 score=0.7760

Xlhu7bkCJOA
sem=0.7760

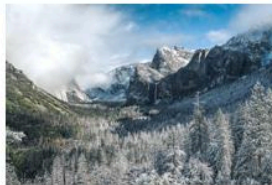
#8 score=0.7707

1RcDcdEs6Ps
sem=0.7707

#9 score=0.7690

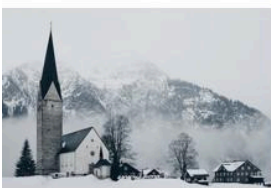
fHK6kIQmtTw
sem=0.7690

#10 score=0.7632

fwxVU3WnsFQ
sem=0.7632

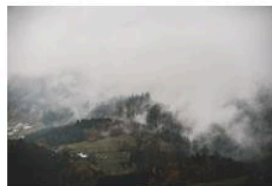
QUERY

9wTWfYlnJ4Y.jpg

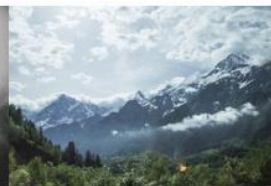


CLIP + STYLE RERANK

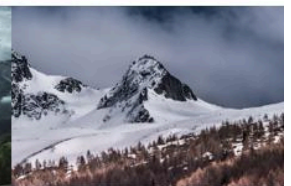
#1 score=0.8495

795KLb6DI7E
sem=0.8292
style=0.9105
final=0.8495

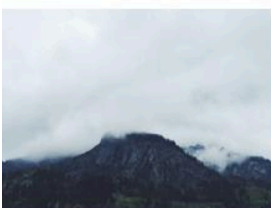
#2 score=0.8230

rZ2LHT8eU9w
sem=0.8116
style=0.8572
final=0.8230

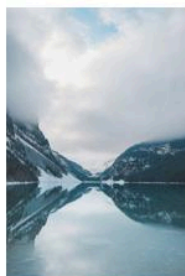
#3 score=0.8121

hB16dqA6uv4
sem=0.8062
style=0.8300
final=0.8121

#4 score=0.8120

cDD83wV627U
sem=0.7873
style=0.8860
final=0.8120

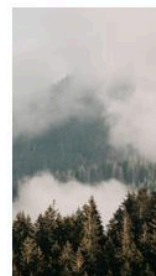
#5 score=0.7964

SZs2dq_Y38w
sem=0.7516
style=0.9309
final=0.7964

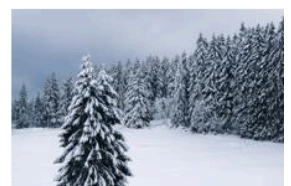
#6 score=0.7943

6xTKw5ufGTy
sem=0.7624
style=0.8901
final=0.7943

#7 score=0.7891

G73eQ6uBoYg
sem=0.7573
style=0.8843
final=0.7891

#8 score=0.7885

OoqKyKdMBXw
sem=0.7624
style=0.8667
final=0.7885

#9 score=0.7866

-87JyMb9ZfU
sem=0.7311

#10 score=0.7864

1RcDcdEs6Ps
sem=0.7707

style=0.9528
final=0.7866



style=0.8334
final=0.7864



Observations

- Forest scenes remain semantically relevant.
- Reranking favors visually similar atmosphere and mood.

Query Image: 39DcBUbYZP4

QUERY

39DcBUbYZP4.jpg



CLIP ONLY

#1 score=0.8230

tQlWjhG1148
sem=0.8230

#2 score=0.8200

5tKA14ysaRs
sem=0.8200

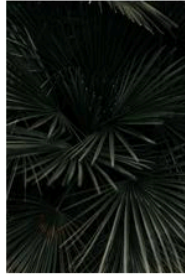
#3 score=0.8012

mFECC6AY67E
sem=0.8012

#4 score=0.7966

-FnGodn23_c
sem=0.7966

#5 score=0.7908

nUrfdD3GQLU
sem=0.7908

#6 score=0.7904

YcCTVdhPiw4
sem=0.7904

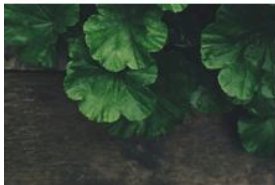
#7 score=0.7878

6XwkG4B5xAU
sem=0.7878

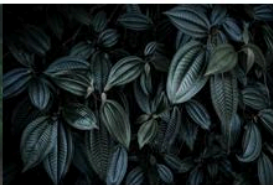
#8 score=0.7856

BAPzw6CKBQY
sem=0.7856

#9 score=0.7853

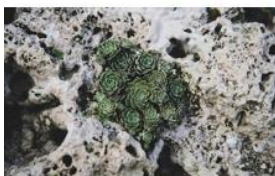
qrCSGTkkDXo
sem=0.7853

#10 score=0.7847

iSdpb3NuNWA
sem=0.7847

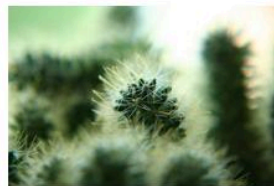
QUERY

39DcBUbYZP4.jpg

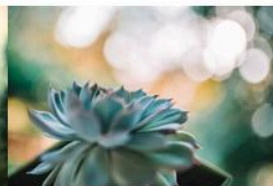


CLIP + STYLE RERANK

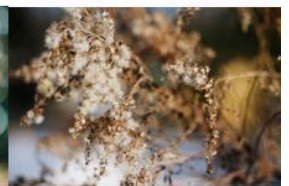
#1 score=0.8430

tQlWjhG1148
sem=0.8230
style=0.9028
final=0.8430

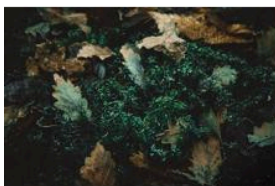
#2 score=0.8278

mFECC6AY67E
sem=0.8012
style=0.9075
final=0.8278

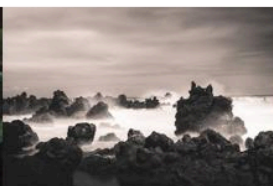
#3 score=0.8161

YcCTVdhPiw4
sem=0.7904
style=0.8931
final=0.8161

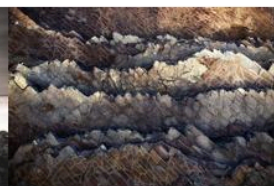
#4 score=0.8066

5tKA14ysaRs
sem=0.8200
style=0.7664
final=0.8066

#5 score=0.8002

8sQoBzPNdpQ
sem=0.7426
style=0.9730
final=0.8002

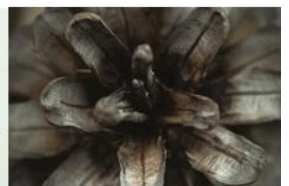
#6 score=0.8002

rUoa7RXJaWI
sem=0.7752
style=0.8751
final=0.8002

#7 score=0.7995

Ji7kcVi8KHs
sem=0.7763
style=0.8691
final=0.7995

#8 score=0.7960

BAPzw6CKBQY
sem=0.7856
style=0.8273
final=0.7960

#9 score=0.7947

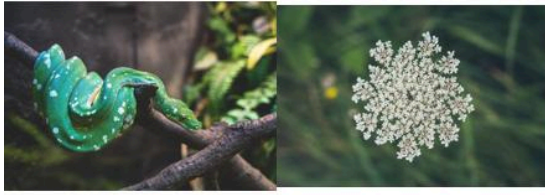
72vLY-QMd3w
sem=0.7717

#10 score=0.7939

ymOROKFI8tE
sem=0.7821

style=0.8637
final=0.7947

style=0.8292
final=0.7939



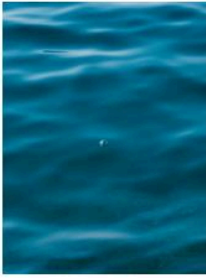
Observations

- Semantic retrieval finds similar content.
- Style reranking improves brightness and color consistency.

Query Image: A-G8q9zorGs

QUERY

A-G8q9zorGs.jpg



CLIP ONLY

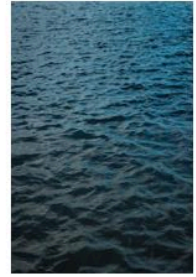
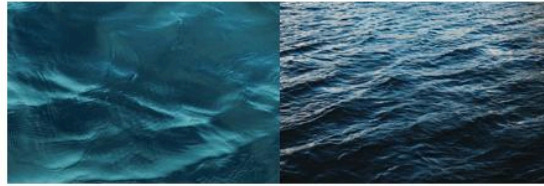
#1 score=0.9184

EWZAHjBzdb8
sem=0.9184

#2 score=0.9103

0PUQAOpLSIQ
sem=0.9103

#3 score=0.8998

iCARCmC4IoQ
sem=0.8998

#4 score=0.8916

1braZySIEKA
sem=0.8916

#5 score=0.8834

mF6nc-b1IWc
sem=0.8834

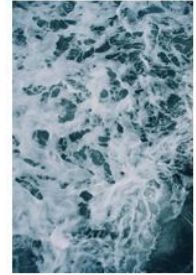
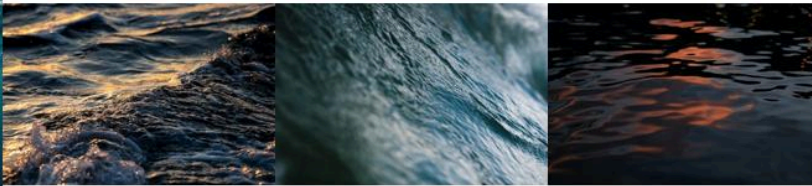
#6 score=0.8834

M_6lB2xafZI
sem=0.8834

#7 score=0.8804

NwH8vv448lY
sem=0.8804

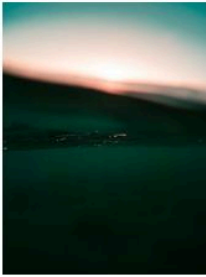
#8 score=0.8778

pO1tGn64oJl
sem=0.8778

#9 score=0.8757

xwTgctQbCtA
sem=0.8757

#10 score=0.8708

ZMEmMguGD1l
sem=0.8708

QUERY

A-G8q9zorGs.jpg



CLIP + STYLE RERANK

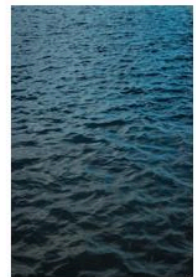
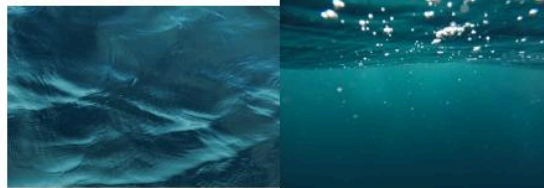
#1 score=0.9032

EWZAHjBzdb8
sem=0.9184
style=0.8574
final=0.9032

#2 score=0.9031

1braZySIEKA
sem=0.8916
style=0.9374
final=0.9031

#3 score=0.8718

iCARCmC4IoQ
sem=0.8998
style=0.7878
final=0.8718

#4 score=0.8678

0PUQAOpLSIQ
sem=0.9103
style=0.7402
final=0.8678

#5 score=0.8522

MmKSSSuvuM
sem=0.8402
style=0.8882
final=0.8522

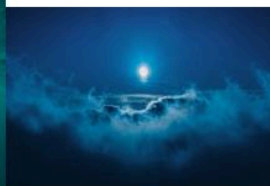
#6 score=0.8480

00taym-piCM
sem=0.8208
style=0.9297
final=0.8480

#7 score=0.8461

cs6Omgf0fU
sem=0.8204
style=0.9233
final=0.8461

#8 score=0.8461

mF6nc-b1IWc
sem=0.8834
style=0.7341
final=0.8461

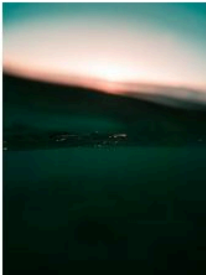
#9 score=0.8410

xwTgctQbCtA
sem=0.8757

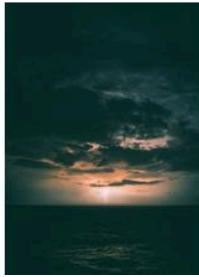
#10 score=0.8374

AKIzQNrtXuW
sem=0.8635

style=0.7369
final=0.8410



style=0.7592
final=0.8374



Observations

- Water-related images remain highly relevant.
- Reranking prefers similar color distributions.

Quantitative Evaluation

Results generated by:

```
python experiments/evaluate_style_reranking.py
```

Evaluation Output

Paste terminal output here:

```
Evaluating reranking on 30 query images

Loading CLIP model on device: cpu
Model loaded
Tokenizer loaded
Loading CLIP model on device: cpu
Model loaded
Tokenizer loaded
CLIP only:
avg brightness difference = 0.1698
avg contrast difference = 0.0745
avg saturation difference = 0.1821
avg warmth difference = 0.0510

Reranked:
avg brightness difference = 0.0924
avg contrast difference = 0.0577
avg saturation difference = 0.1239
avg warmth difference = 0.0399

Improvement (positive means reranked is closer in style):
brightness: 0.0774
contrast: 0.0167
saturation: 0.0582
```

```
warmth: 0.0111
```

Discussion

Advantages:

- Strong semantic retrieval through CLIP.
- Improved style consistency after reranking.
- Modular architecture.
- Easily scalable to larger datasets.

Limitations:

- Style descriptors are handcrafted.
- Color histogram is relatively simple.
- Current dataset contains only 1000 images.
- Text queries currently do not use style reranking.

Future Work

Planned improvements:

1. Scale dataset to 25,000 images.
2. Build HNSW index.
3. Build IVF index.
4. Benchmark retrieval speed and recall.
5. Generate synthetic style variations.
6. Deploy FastAPI backend.
7. Create web interface.

Conclusion

The project successfully combines:

```
CLIP Semantic Retrieval  
+  
Photographic Style Reranking
```

The resulting system retrieves images that are closer to the query not only semantically but also visually, making it more suitable for photographer-oriented search applications.